A number saves no one. A warning people act on does.

1 · Problem & users

- Task: predict daily river discharge (cfs) per station from rain, temperature, soil moisture + 4 catchment attributes
- No past discharge in test.csv: the river's 'memory' must come from weather history
- District disaster officer: which rivers need attention this week, and how sure are we?
- Riverside resident: am I safe, what should I do, in my language (English, Hindi, Kannada)
- Evidence: in a Bihar study of 810 households, getting an alert doubled the chance of protective action
- So we built both: an accurate model and an early-warning product on top of it

Data: the three sources from the brief

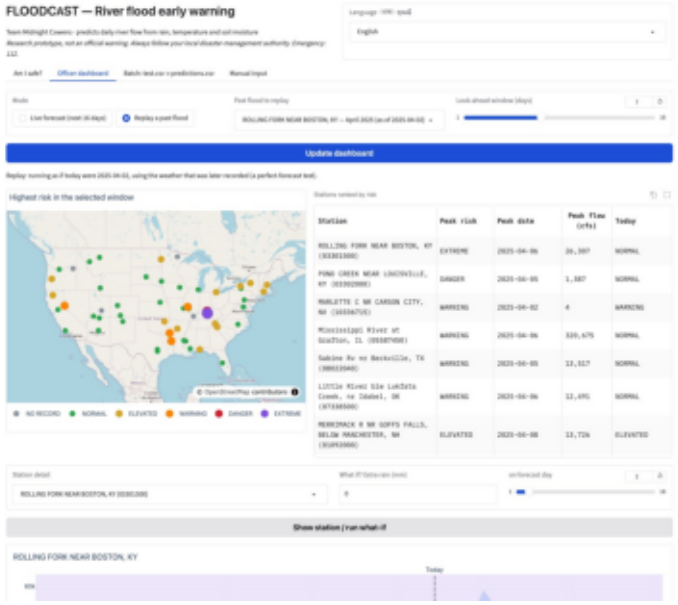
- USGS NWIS daily discharge (param 00060): 49 stations incl. the 5 recommended + big rivers
- Open-Meteo archive at each gauge: precipitation, Tmax, Tmin, soil moisture 0–7 cm (m^3/m^3)
- Catchment attributes: USGS GAGES-II (StreamStats API was offline): area, elevation, slope, forest
- Train 2012–2022 (193,438 station-days) | held-out test 2023-01 → 2025-10 (49,361 station-days), never used for any
- Cleaning: drop ice/equipment-affected days and USGS no-data codes; never fill missing targets
- Live product: Open-Meteo 16-day forecast (free, cached, offline snapshot fallback)

Model + product

3 · CPU LightGBM ensembles → risk levels → alerts

- Features: rolling rain 3–90 d, antecedent-precipitation index, soil trend, snow/melt, season
- Known stations: 3-objective ensemble + station climatology; unseen: flow per sq mi × area
- Risk levels per river: top-10% day, 2-, 5-, 20-year flood (Gumbel), bias-corrected (GloFAS method)
- Likely range (split-conformal), plain-language 'why' (SHAP), what-if rainfall, CAP 1.2 export
- Fine-tuned for the hidden test: x1.25 median-to-mean fix (new rivers), cold-start rain fill, capacity search
- Final model retrained on all data 2012 to Oct 2025; offline on an 8 GB laptop; deterministic; 71 tests





Results on the 2023–2025 holdout

4 · Accuracy (raw cfs, evaluation model trained 2012–2022) and warning skill

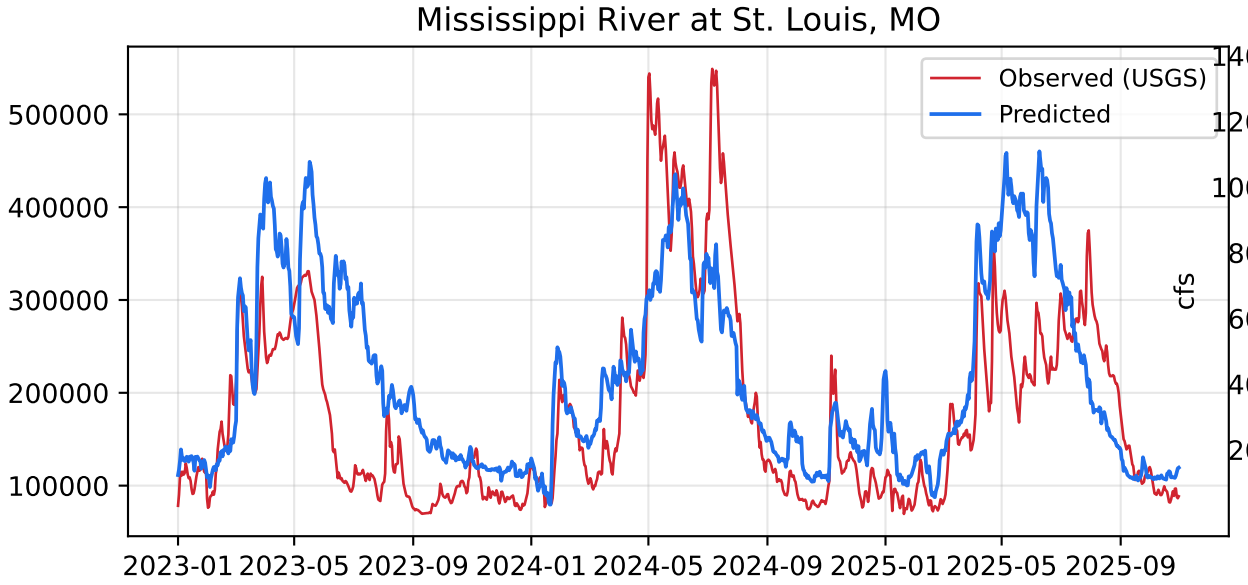
	RMSE	RMSE top-10%	MAE	R^2
Monthly climatology baseline	45,913	50,655	12,116	0.815
Ours — known stations	38,807	55,561	8,914	0.867
Ours — unseen stations (CV)	54,290	116,158	15,087	0.741

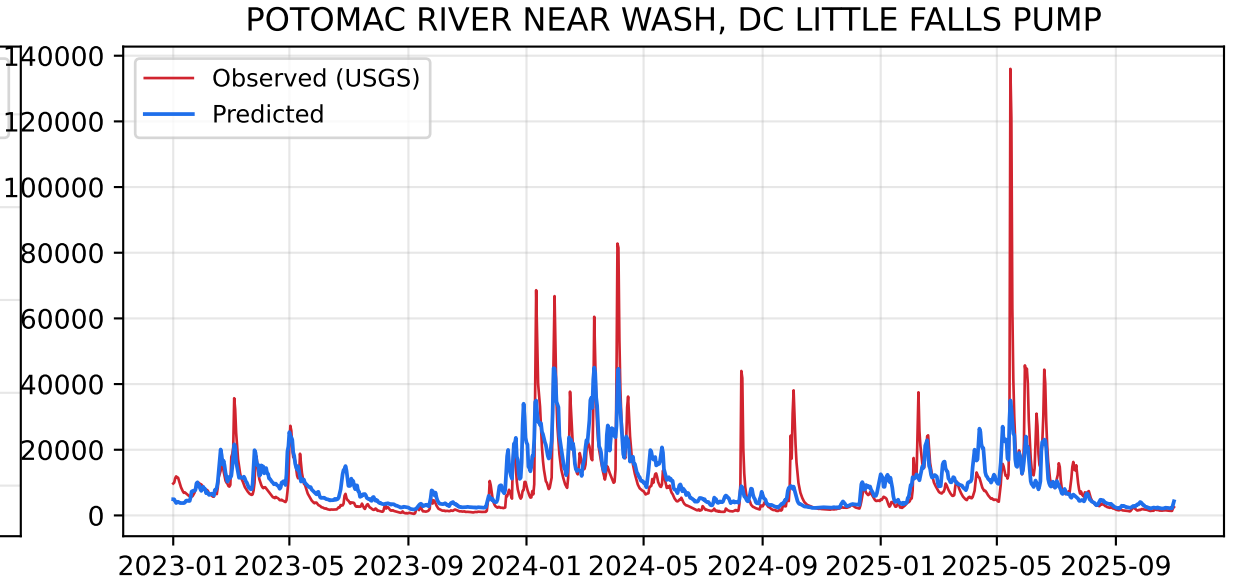
Alert level	Caught (POD)	False alarms
Elevated	66%	33%
Warning	36%	55%
Danger	16%	59%

Warning skill uses a ±2-day timing tolerance · exact level right on 91% of days, within one level 99%.

Likely range covers 72% of observations (target 80%). Replay: Rolling Fork, KY flagged EXTREME 4 days ahead of the April 2025 flood.

Fine-tuning (2 time folds, files starting cold like a judge's test.csv): new rivers RMSE -9% to -15%, flood-peak RMSE -17% to -29%, R^2 0.73-0.75 -> 0.78-0.82.





Limitations & responsible use

5 · Limitations

- Big regulated rivers (Mississippi, Ohio) follow upstream rain and dam releases, not gauge weather
- Peaks are under-predicted in magnitude; risk levels correct for this, but rare 20-year floods are mostly missed
- Dam-controlled rivers (e.g. Colorado at Austin) are poorly captured day-to-day
- Likely range is too narrow in 2023–25 (72% vs 80% target): a shift from the training years
- Replay uses recorded weather (perfect forecast); real forecasts add error, more so on days 8–16
- Hindi/Kannada text is machine-assisted: native review before real use. Research prototype; dial 112